

Unit 6 Vocabulary



Work smarter

Inglés Profesional





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6.1.

Telecommunications

Telecommunications (or telecom) refers to the transmission of information over long distances using electronic methods. It encompasses voice, video, and data communication, and includes a wide range of modern technologies and infrastructures.

These technologies allow us to stay connected—whether it's making a phone call, sending an email, streaming a video, or navigating with GPS. The field of telecommunications plays a central role in our daily lives and in global connectivity.

6.2.

Main forms of Telecommunication

- > **Wired Communication** uses physical mediums such as copper wires or fiber-optic cables. Examples: landline phones, broadband internet, Ethernet networks.
- > **Wireless Communication:** transmits data without wires via electromagnetic waves. Examples: Wi-Fi, mobile networks (3G/4G/5G), Bluetooth, satellite communications.

6.3.

Telecommunication circuits

A **telecommunication** circuit typically includes:

- > **Two stations**, each with:
 - » A transmitter: sends the signal.
 - » A receiver: receives the signal.
 - » Or a transceiver: a device that performs both functions.
- > The **medium** (channel for transmission) can be:
 - » Wire (copper)
 - » Optical fiber
 - » Electromagnetic waves (used in wireless systems)
 - » Light



6.4.

Antennae (Aerials)

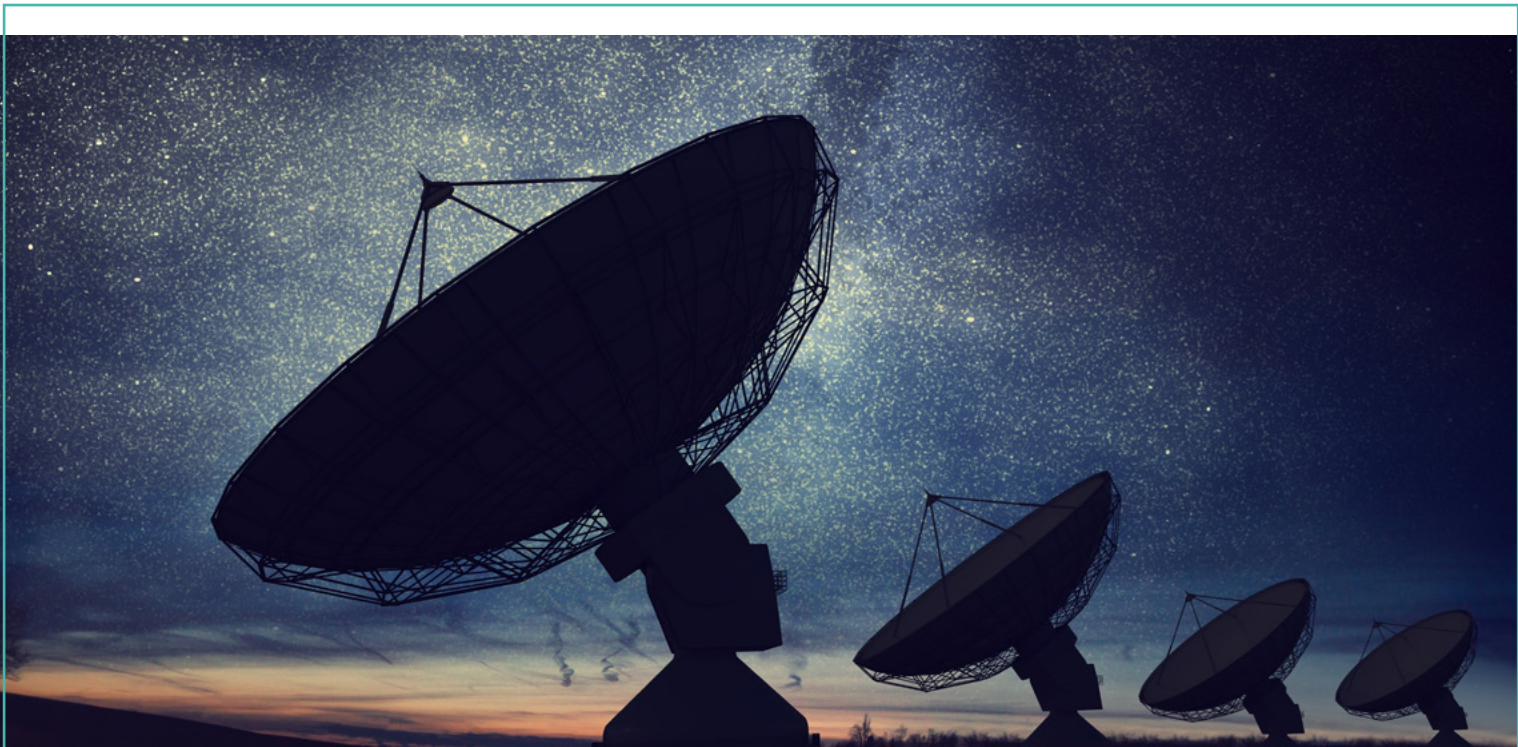
An **antenna** or **aerial** is a device that radiates or receives radio waves. It plays a key role in wireless communication systems.

Common antenna types include:

- > **Dipole Antenna:** basic design for radio transmission.
- > **Yagi Antenna:** directional, used in TV and amateur radio.
- > **Loop Antenna:** circular design, used in low-frequency applications.
- > **Fractal Antenna:** compact, complex design for multi-frequency use.
- > **Satellite Dish:** parabolic antenna for satellite TV or communication.

Antennas are found in:

- > **Televisions**
- > **Radios**
- > **Mobile phones**
- > **Wi-Fi routers**
- > **Satellites**
- > **Radar systems**





6.5.

Let's practise!

1. Label the pictures below with their appropriate name.

Satellite dish

Dipole antenna

Yagi antenna

Loop antenna

Fractal antenna



1.



2.



3.



4.



5.

2. Match the following terms (A-E) with their correct definitions (1-5)

1. A metal dish that receives signals from orbiting space objects
2. Communication method without physical cables
3. Device that both transmits and receives data
4. Cable system that uses light to transmit information
5. Circular antenna used for receiving low-frequency waves

- a. Transceiver
- b. Wireless
- c. Fibre optics
- d. Satellite dish
- e. Loop antenna



6.6.

Test yourself!

1. What does "telecommunications" refer to?
 - a. Only TV and radio broadcasting
 - b. The use of satellites only
 - c. Computer programming
 - d. The transmission of information over a distance
2. Which device combines both a transmitter and a receiver?
 - a. Transceiver
 - b. Amplifier
 - c. Antenna
 - d. Decoder
3. Which of these is a wired form of communication?
 - a. Bluetooth
 - b. Satellite dish
 - c. Optical fibre
 - d. Wi-Fi
4. Which antenna is commonly used for satellite communication?
 - a. Dipole antenna
 - b. Loop antenna
 - c. Fractal antenna
 - d. Satellite dish
5. What is broadcasting?
 - a. Sending messages through cables
 - b. Transmitting content to a wide audience via radio or TV
 - c. Private messaging
 - d. Encrypting information
6. What is the function of a Yagi antenna?
 - a. It's omnidirectional
 - b. It's used to clean signals
 - c. It sends Wi-Fi signals
 - d. It's directional and used for receiving TV signals
7. Which term refers to communication without cables?
 - a. Wireless
 - b. Broadcasting
 - c. Wired
 - d. Loop transmission



8. Which technology uses light to transmit data?
 - a. Copper cable
 - b. Optical fibre
 - c. Transceiver
 - d. Wi-Fi
9. What type of antenna is suitable for low-frequency transmission?
 - a. Satellite dish
 - b. Dipole antenna
 - c. Loop antenna
 - d. Yagi antenna
10. Fractal antennas are used because they are:
 - a. Easy to build manually
 - b. Good for low-frequency only
 - c. Large and heavy
 - d. Small, complex, and support multiple frequencies





6.7.

Answer key

Let's practise!

1.

1. Yagi antenna
2. Satellite Dish
3. Loop
4. Dipole antenna
5. Fractal antenna

2.

1. d)
2. b)
3. a)
4. c)
5. e)

Test yourself - Answer key

- | | | |
|------|------|-------|
| 1. D | 5. B | 9. C |
| 2. A | 6. D | 10. D |
| 3. C | 7. A | |
| 4. D | 8. B | |



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